# Homework 7

Modules 7b–8 · Homework · No date

Name _____ Date _____

1. A 75-kg volunteer received a single IV bolus dose of an investigational drug. The drug was administered at a dose of 15 mg/kg. The drug has an apparent volume of distribution that is 0.3 L/kg and an elimination half-life approximately 4 hours .

**1a** Module 7b

What is the AUC following the IV bolus dose?

**1b** Module 7b

If the MEC is 3.75 mg/L, what is the duration of activity of this drug in this volunteer?

**1c** Module 7b

What is the duration of activity if the dose were 7.5 mg/kg?

**1d** Module 7b

What dose would provide a duration of action of 8 hours?

2. A 68-kg, 28-year-old female is receiving phenytoin for the treatment of seizures. When this patient was taking a daily oral dose of 300 mg, her steady state plasma concentration was 9.12 mg/L. The daily dose was increased to 400 mg which resulted

in a steady-state plasma concentration of 21.8 mg/L. Assume 100% bioavailability and an apparent volume of distribution that is equal to 70% of body weight.

**2a** Module 8

What is this patient's KM?

**2b** Module 8

What is the Vmax in this patient?

**2c** Module 8

What is the daily dose rate of phenytoin that should achieve a steady state phenytoin plasma concentration of 15 mg/L in this patient? (Round to the nearest ten)

**2d** Module 8

What is the half-life and clearance in this patient while she is taking the dose that you recommended above to achieve a steady state concentration of 15 mg/L? What is the half-life?

**2d** Module 8

What is the half-life and clearance in this patient while she is taking the dose that you recommended above to achieve a steady state concentration of 15 mg/L? What is the clearance?

**2e** Module 8

What is the expected steady state concentration if this patient were to take 325 mg/day?

# Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

## 1a

Worked by the drill

**Answer:** 288.6 (mg/L)·hr

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **weight** | 75-kg | scales the dose and the volume |
| **$D_0$** | 15 mg/kg | × 75 kg = 1125 mg |
| **$V_D$** | 0.3 L/kg | × 75 kg = 22.5 L |
| **t½** | approximately 4 hours | k = 0.17325 $hr^{-1}$ for Cl = $kV_D$ |

### THE WORKING, ONE LINE AT A TIME

UNIT

$$D_0 = 15 \text{ mg/kg} \times 75 \text{ kg} = 1125 \text{ mg};\ V_D = 0.3 \text{ L/kg} \times 75 \text{ kg} = 22.5 \text{ L}$$

Dose and volume are both per kilogram, so each is multiplied by the 75-kg weight; the kilograms cancel and leave mg and L.

SETUP

$$k = \frac{0.693}{4 \text{ hr}} = 0.17325 \text{ hr}^{-1};\ Cl_T = k \times V_D = 0.17325 \times 22.5 \text{ L} = 3.898 \text{ L/hr}$$

Clearance is not given, so it is built from the half-life and the volume: $Cl = kV_D$, total body clearance. $hr^{-1}$ × L leaves L/hr.

ALGEBRA

$$AUC = \frac{F\ D_0}{Cl_T} = \frac{1125 \text{ mg}}{3.898 \text{ L/hr}} = 288.6 \text{ (mg/L)·hr}$$

AUC, the area under the concentration–time curve, from $Cl = \frac{F\ D_0}{AUC}$ rearranged. F = 1 for an IV bolus, because the whole dose goes directly into the blood. mg over L/hr leaves (mg/L)·hr.

ROUND

$$AUC = 288.6 \text{ (mg/L)·hr} = \frac{C_0}{k} = \frac{50 \text{ mg/L}}{0.17325 \text{ hr}^{-1}}$$

The AUC is the total exposure from this one dose, the quantity an exposure–response relationship sets against the effect.

> **DOES THE NUMBER MAKE SENSE?**
> AUC = $C_0 \div k$, with $C_0 = 1125 \div 22.5 = 50$ mg/L and k = 0.17325 $hr^{-1}$: 288.6 (mg/L)·hr.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

---

## 1b

Worked by the drill

**Answer:** 14.95 hr

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **Ceff** | 3.75 mg/L | the MEC, where the duration ends |
| **$D_0$** | 15 mg/kg | over 0.3 L/kg gives $C_0$ = 50 mg/L |
| **$V_D$** | 0.3 L/kg | the denominator of $C_0$ |
| **t½** | approximately 4 hours | k = 0.17325 $hr^{-1}$ |
| **weight** | 75-kg | cancels in $C_0$ |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$C_0 = \frac{D_0}{V_D} = \frac{15\ mg/kg}{0.3\ L/kg} = 50\ mg/L;\ k = \frac{0.693}{4\ hr} = 0.17325\ hr^{-1}$$

Dose and volume are both per kilogram, so the kilograms cancel in $C_0$: 1125 mg over 22.5 L is the same 50 mg/L. k comes from the half-life.

ALGEBRA

$$teff = \frac{\ln C_0 - \ln Ceff}{k} = \frac{\ln(50 \div 3.75)}{0.17325\ hr^{-1}} = \frac{\ln 13.33}{0.17325} = \frac{2.590}{0.17325\ hr^{-1}}$$

The duration is the time the level takes to fall from $C_0$ to the MEC, the minimum effective concentration; the logarithm of their ratio over k gives that time.

ROUND

$$teff = 14.95\ hr$$

2.590 ÷ 0.17325 = 14.95 hr: between 3 and 4 half-lives of 4 hr, because the 13.33-fold fall lies between 8 and 16.

> **DOES THE NUMBER MAKE SENSE?**
> $C_0$ ÷ Ceff = 13.33 lies between 8 and 16, so teff is between 3 and 4 half-lives of 4 hr: 14.95 hr.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

---

## 1c

Worked by the drill

**Answer:** 10.95 hr

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **$D_0$** | 7.5 mg/kg | half of 15 mg/kg, so $C_0$ halves to 25 mg/L |
| **Ceff** | 3.75 mg/L | the MEC from part b |
| **$V_D$** | 0.3 L/kg | unchanged by the dose |
| **t½** | approximately 4 hours | the amount teff falls by |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$C_0 = \frac{7.5\ mg/kg}{0.3\ L/kg} = 25\ mg/L;\ k = 0.17325\ hr^{-1};\ Ceff = 3.75\ mg/L$$

Half the dose gives half of $C_0$, because the volume does not change with the dose; k and the MEC are the same as in part b.

ALGEBRA

$$\text{teff} = \frac{\ln(25 \div 3.75)}{0.17325\ \text{hr}^{-1}} = \frac{\ln 6.667}{0.17325} = \frac{1.897}{0.17325\ \text{hr}^{-1}} = 10.95\ \text{hr}$$

The level now has to fall only 6.667-fold, from 25 to 3.75 mg/L, to reach the MEC, instead of 13.33-fold, so the duration is shorter than in part b.

ROUND

$14.95 - 10.95 = 4$ hr, one half-life; teff = 10.95 hr

Halving the dose removes one doubling, and each doubling adds one half-life, so 4 hr comes off the duration; the 14.95 hr is not halved.

**DOES THE NUMBER MAKE SENSE?**

Half the dose removes one half-life, 4 hr, from 14.95 hr: 10.95 hr, not half of 14.95.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

---

## 1d

Worked by the drill

**Answer:** 337.4 mg

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **teff** | 8 hours | the target duration, in the exponent |
| **Ceff** | 3.75 mg/L | the MEC from part b |
| **$V_D$** | 0.3 L/kg | × 75 kg = 22.5 L |
| **t½** | approximately 4 hours | k = 0.17325 $\text{hr}^{-1}$ |
| **weight** | 75-kg | scales the volume |

### THE WORKING, ONE LINE AT A TIME

UNIT

$$V_D = 0.3\ \text{L/kg} \times 75\ \text{kg} = 22.5\ \text{L};\ k = \frac{0.693}{4\ \text{hr}} = 0.17325\ \text{hr}^{-1}$$

The dose line wants the patient's volume in L, so the per-kilogram volume is multiplied by 75 kg; the half-life gives k for the exponent.

ALGEBRA

$$D_0 = \text{Ceff} \times V_D \times e^{k\ \text{teff}} = (3.75\ \text{mg/L})(22.5\ \text{L})\ e^{(0.17325)(8)} = 84.38\ \text{mg} \times e^{1.386}$$

$C = C_0 e^{-kt}$ with $C_0 = D_0$ over $V_D$, solved for the dose at the moment the level reaches the MEC. mg/L × L leaves 84.38 mg, the amount in the body at the MEC.

ALGEBRA

$e^{1.386} = 3.999$, so $D_0 = 84.38\ \text{mg} \times 3.999 = 337.4\ \text{mg}$

8 hours is two half-lives of 4 hr, so the level falls fourfold over that time and the dose must be four times the 84.38 mg at the MEC.

ROUND

$D_0 = 337.4$ mg (4.50 mg/kg for 75 kg)

337.4 mg, or 4.50 mg/kg in the per-kilogram form the stem doses in; a lower dose than the 15 mg/kg, because 8 hours is shorter than the duration at 15 mg/kg.

**DOES THE NUMBER MAKE SENSE?**

8 hours is two half-lives of 4 hr, so the dose is 4 times the 84.38 mg at the MEC: 337.4 mg.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

---

## 2a

Worked by the drill

**Answer:** 6.87 mg/L

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **R1** | 300 mg | the first daily dose |
| **$C_1$** | 9.12 mg/L | steady state on 300 mg |
| **R2** | 400 mg | the increased daily dose |
| **$C_2$** | 21.8 mg/L | steady state on 400 mg |
| **weight** | 68-kg | *not needed for KM* |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$KM = \frac{R2 - R1}{(R1/C_1) - (R2/C_2)}$$, with R1 = 300 mg/day, $C_1$ = 9.12 mg/L, R2 = 400 mg/day, $C_2$ = 21.8 mg/L

Two daily doses in the same patient and the steady-state concentration each produced: the direct method for KM.

ALGEBRA

$$KM = \frac{400 - 300}{(300/9.12) - (400/21.8)} = \frac{100}{32.895 - 18.349} = \frac{100}{14.546}$$

Each dose over its own concentration, carried to three decimals because the two ratios are close and rounding them early moves KM noticeably.

ROUND

KM = 6.87 mg/L

(mg/day) over (mg/day per mg/L) leaves mg/L, a concentration. The unrounded 6.875 mg/L is carried into Vmax and the later parts.

**DOES THE NUMBER MAKE SENSE?**

A one-third rise in dose more than doubled the concentration, so the drug is near saturation; KM is a concentration: 6.87 mg/L.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

---

## 2b

Worked by the drill

**Answer:** 526.1 mg/day

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **KM** | 6.875 mg/L | from part a |

| | | |
|---|---|---|
| **R2** | 400 mg | the dose used |
| **$C_2$** | 21.8 mg/L | its own steady state |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$\text{Vmax} = \frac{R(KM + C_{ss})}{C_{ss}}$$

The dose-rate line solved for Vmax: multiply both sides by (KM + $C_{ss}$) and divide by $C_{ss}$, using one dose and the steady-state concentration it produced.

ALGEBRA

$$\text{Vmax} = \frac{(400\ \text{mg/day})(6.875 + 21.8)\ \text{mg/L}}{21.8\ \text{mg/L}} = \frac{11470}{21.8} = 526.1\ \text{mg/day}$$

mg/L cancels and Vmax keeps the units of the dose, mg/day. The 300 mg/day with 9.12 mg/L pair gives 526.1 mg/day as well.

ROUND

$$\text{Vmax} = 526.1\ \text{mg/day}$$

526.14 mg/day to one decimal place. It is above the 400 mg/day dose, as it must be for a dose that reached a steady state.

> **DOES THE NUMBER MAKE SENSE?**
> Vmax must exceed the 400 mg/day dose that reached a steady state: 526.1 mg/day.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

---

## 2c Worked by the drill

**Answer:** 360 mg/day

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **KM** | 6.875 mg/L | from part a |
| **Vmax** | 526.1 mg/day | from part b |
| **$C_{ss}$** | 15 mg/L | the target |
| **rounding** | nearest ten | 360.8 becomes 360 |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$R = \frac{\text{Vmax} \times C_{ss}}{KM + C_{ss}}$$

The patient's own KM and Vmax, from parts a and b, with the target of 15 mg/L, in the nonlinear dose-rate line.

ALGEBRA

$$R = \frac{(526.1\ \text{mg/day})(15\ \text{mg/L})}{(6.875 + 15)\ \text{mg/L}} = \frac{7891.5}{21.875} = 360.8\ \text{mg/day}$$

mg/L cancels and leaves mg/day. 15 mg/L lies between 9.12 and 21.8 mg/L, so the dose lies between 300 and 400 mg/day.

ROUND

$R = 360$ mg/day

The stem asks for the nearest ten, so 360.8 mg/day is reported as 360 mg/day, a dose that can be given and that keeps $C_{ss}$ near 15 mg/L.

**DOES THE NUMBER MAKE SENSE?**

15 mg/L lies between 9.12 and 21.8 mg/L, so the dose lies between 300 and 400 mg/day: 360 mg/day.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

---

## 2d

Worked by the drill

**Answer:** 1.37 days

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **weight** | 68-kg | $V_D = 0.7 \times 68 = 47.6$ L |
| **$V_D$** | 70% of body weight | 0.7 L/kg |
| **$C_p$** | 15 mg/L | the concentration the dose achieves |
| **KM** | 6.875 mg/L | from part a |
| **Vmax** | 526.1 mg/day | from part b |

### THE WORKING, ONE LINE AT A TIME

UNIT

$V_D = 0.70 \times 68$ kg $= 47.6$ L

"70% of body weight" is read as 0.7 L/kg, taking 1 L as 1 kg, so the volume is 0.7 times the 68 kg weight, in litres.

SETUP

$t½ = V_D(\frac{0.693}{Vmax})(KM + C_p)$, at $C_p = 15$ mg/L

The recommended dose was chosen to give 15 mg/L, so the half-life is calculated at that concentration; at another concentration it would differ.

ALGEBRA

$$t½ = 47.6 \text{ L} \times \frac{0.693}{526.1 \text{ mg/day}} \times (6.875 + 15) \text{ mg/L} = 1.372 \text{ days}$$

L × mg/L gives mg, which cancels against mg/day and leaves days. Using the volume as 0.7 gives 0.02, a result per kilogram.

ROUND

$t½ = 1.37$ days

1.372 days to two decimal places. The rounded 360 mg/day dose gives 14.9 mg/L and 1.36 days, within the same tolerance.

**DOES THE NUMBER MAKE SENSE?**

Vmax is in mg/day, so the half-life comes out in days: 1.372 days, more than one day and less than two.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

---

## 2d

Worked by the drill

**Answer:** 24.05 L/day

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **$C_p$** | 15 mg/L | added to KM in the denominator |
| **KM** | 6.875 mg/L | from part a |
| **Vmax** | 526.1 mg/day | from part b |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_T = \frac{Vmax}{KM + C_p}, \text{ at } C_p = 15 \text{ mg/L}$$

The recommended dose was chosen to give 15 mg/L, and the clearance of a saturable drug is calculated at its concentration.

ALGEBRA

$$Cl_T = \frac{526.1 \text{ mg/day}}{(6.875 + 15) \text{ mg/L}} = \frac{526.1}{21.875} = 24.05 \text{ L/day}$$

mg/day over mg/L leaves L/day, a volume of plasma cleared per unit time. The volume of distribution is not needed for clearance.

ROUND

$$Cl_T = 24.05 \text{ L/day}$$

To two decimal places. A higher dose would raise $C_p$ and lower this clearance, because $C_p$ is in the denominator.

> **DOES THE NUMBER MAKE SENSE?**
> Clearance at 15 mg/L is well below Vmax ÷ KM, because KM + $C_p$, 21.875 mg/L, is larger than KM alone: 24.05 L/day.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.

---

## 2e

Worked by the drill

**Answer:** 11.11 mg/L

### WHAT THE STEM GIVES

| | | |
|---|---|---|
| **R** | 325 mg/day | the new dose rate |
| **KM** | 6.875 mg/L | from part a |
| **Vmax** | 526.1 mg/day | from part b |

### THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{ss} = \frac{R \times KM}{Vmax - R}$$

The dose-rate line solved for $C_{ss}$: multiplying out gives $R \times KM + R \times C_{ss} = Vmax \times C_{ss}$, so $R \times KM = C_{ss}(Vmax - R)$, then divide by $(Vmax - R)$.

ALGEBRA

$$C_{ss} = \frac{(325 \text{ mg/day})(6.875 \text{ mg/L})}{(526.1 - 325) \text{ mg/day}} = \frac{2234.4}{201.1} = 11.11 \text{ mg/L}$$

mg/day cancels and leaves mg/L. The denominator is Vmax minus the dose; adding them gives 2.6 mg/L, below the 9.12 mg/L that 300 mg produced.

ROUND

$$C_{ss} = 11.11 \text{ mg/L}$$

11.11 mg/L to two decimal places. 325 mg/day is 8% above 300 mg/day, yet $C_{ss}$ rises 22% from 9.12 mg/L: the steep rise of a saturable drug.

**DOES THE NUMBER MAKE SENSE?**

325 mg/day is above the 300 mg/day that gave 9.12 mg/L and below the dose that gives 15 mg/L, so $C_{ss}$ lies between them: 11.11 mg/L.

Her homework with her own numbers. She gave no key for it, so this answer is computed from her stem.